

# Johan MAZOYER

Research Interests: Optical Instrumentation, Exoplanets, Debris Disks, Direct Imaging & Coronagraphy, Observation & Characterization of Extrasolar Systems

## 1 RESEARCH POSITIONS

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| CNRS Scientist – <b>LIRA/Paris Observatory - PSL</b> (France)            | Since 2020  |
| Carl Sagan Fellow – <b>NASA Jet Propulsion Laboratory</b> (Pasadena, CA) | 2018 - 2019 |
| Postdoc – <b>Johns Hopkins University</b> (Baltimore, MD)                | 2016 - 2018 |
| Postdoc – <b>Space Telescope Science Institute</b> (Baltimore, MD)       | 2014 - 2016 |
| Graduate Student – <b>LIRA/Paris Observatory - PSL</b> (France)          | 2011 - 2014 |

## 2 EDUCATION

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| HDR (French habilitation) – <b>Paris Observatory - PSL</b> (France)                                                                                                                            | March 2024 |
| PhD – Astronomy & Astrophysics – <b>Université Paris Cité</b> (Paris, France)<br><i>Thesis: High-Contrast Imaging Of Exoplanets And Circumstellar Disks (P. Baudoz &amp; G. Rousset)</i>       | Sept. 2014 |
| Master – Astrophysics & Planetology – <b>Université de Toulouse</b> (Toulouse, France)<br><i>Thesis: Influence of Mars atmosphere on ChemCam detection limits (O. Gasnault &amp; R. Wiens)</i> | Sept. 2011 |
| Master – Techniques for Space Imaging – <b>ISAE Supaero</b> (Toulouse, France)                                                                                                                 | Sept. 2011 |
| Bachelor – Embedded systems – <b>Ecole polytechnique</b> (Paris, France)                                                                                                                       | Sept. 2010 |

## 3 GRANTS & AWARDS

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| ERC - Consolidator Grant (PI) <b>ECHOES</b> - 2 M€                                          | Since 2026 |
| ANR / PEPR Origins (PI) <b>AMINO</b> - 1.2 M€                                               | Since 2026 |
| ANR JCJC (PI) (terminated and reimbursed) - 370 k€                                          | 2025-2026  |
| <b>Académie spatiale d'île de France</b> (PI) 1/2 PhD grant fund for L. Delaye - 60 k€/3yrs | 2025       |
| CNES (PI) ( <b>French space agency</b> ) 1/2 PhD grant for M. Castañeda Medina - 60 k€/3yrs | 2025       |
| DIM Origins (PI) Hardware funding for a spatial light modulator - 12 k€                     | 2023       |
| CNES (co-PI) ( <b>French space agency</b> ) Postdoc fund for I. Luginja - 120 k€/2 yrs      | 2022       |
| Data Intensive Artificial Intelligence program (PI) 1/2 PhD grant Y. Gutierrez - 60k€/3yrs  | 2021       |
| <b>Ecosud</b> Program (PI) Collaboration <i>Universidad de Chile</i> – 50 k€                | 2020       |
| NASA Group Award: LBTI Hosts Survey Science Team                                            | 2020       |
| Carl Sagan Fellowship ( <b>NASA Hubble Fellowship Program</b> ) – 280k€/3 yrs               | 2018       |
| Cover of <b>Astronomy &amp; Astrophysics</b> Journal ( <b>Volume 564</b> )                  | 2014       |
| Thesis Award (CNES fellow symposium JC2)                                                    | 2013       |

## 4 OUTREACH



I am heavily involved in scientific outreach in France. In addition to regular class or public interventions, I am running institutional events and associative initiatives:

- **Podcast Science:** I am running **PodcastScience.fm**, a **general science program**, airing every weeks, in french. Listened by 10'000 to 20'000 listeners, Podcast Science received the Golden blog award for best scientific blog in 2012.

- **Les p'tits cueilleurs d'étoiles:** Association organizing astronomers visits in children hospitals in France. I am organizing all Paris region hospitals (25 visits/yr).

- **Fête de la science:** I was the main organizer of the annual Paris Observatory Doors Open Days (~1000 visitors each year) for two consecutive years (2023 and 2024)

## 5 PROFESSIONAL ACTIVITIES & SERVICE

### Responsibilities in scientific instruments:

- **Roman Space Telescope Coronagraph:** Deputy CNES representative Since 2023
- **VLT/SPHERE+:** Dark-Hole work package leader Since 2022
- **Habitable Exoplanet Observatory (HabEx):** Contributing Scientist 2019
- **Large UV Optical Infrared Surveyor (LUVOIR):** Contributing Scientist 2019
- **Gemini Planet Imager (GPI):** junior member of the consortium 2017-2020

### Organization of Scientific Meetings:

- HWO 2026 Annual Symposium (LOC) Massy, 2026
- Roman coronagraphic instrument summer school (SOC) Nice, March 2026
- ExoSystèmes 4 (SOC) Lyon, 2024
- National Capital Area Disks conference (SOC & LOC) Baltimore 2018
- Optimal Optical Coronagraphs workshop (SOC & LOC) Leiden, 2017
- High Contrast Imaging from Space (SOC) Baltimore, 2016

### Other services:

- **"Exoplanetary systems" Group Leader** at Paris Observatory / LIRA Since 2025
- **Hubble Telescope Allocation Committee** 2024
- Science Commity of CNRS/INSU's exoplanets group (**CET exoplanètes**) 2023 - 2024
- Science Commity of CNRS/INSU's High Angular Resolution Group (**CS-ASHRA**) Since 2021
- **Reviewer** for *AJ*, *A&A*, *MNRAS*, *PASP* and *JATIS*.
- **External Reviewer** for 2 Phd Committees (France, Belgium).

## 6 MENTORING

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|-------------------------------------------------------------------------------------------|------------|
| Manuela Castañeda Medina (PhD, CNES/ONERA): co-direction with L. Mugnier                  | Since 2026 |
| Lukas Delaye (PhD, LIRA): co-director with A. Potier                                      | Since 2025 |
| Yann Gutierrez (PhD, LIRA): co-director with L. Mugnier, ONERA                            | 2022-2025  |
| Vito Squicciarini (Postdoc, LIRA): co-advisor with A.-M. Lagrange                         | 2022-2024  |
| Iva Leginja (Postdoc, LIRA): main advisor ; CNES Fellow. Now Resercher at Obs Côte D'Azur | 2022-2024  |
| Sophia Stasevic (PhD, LIRA) co-director with A.-M. Lagrange and J. Milli                  | 2021-2025  |
| Justin Hom (PhD, ASU) co-advisor with J. Patience                                         | 2019-2023  |
| Kevin Fogarty (PhD, JHU) co-advisor with L. Pueyo. Now Research Scientist at AMES         | 2017-2019  |

## 7 TEACHING

### Observatoire de Paris Master Class:

- Instrumentation for Astronomy
- Detection of Exoplanets

## RESEARCH ACHIEVEMENTS

I have a longstanding interest in developing innovative high-contrast techniques, which are critical for detecting and characterizing faint objects orbiting closely to bright stars (exoplanets and dust belts). Over the years, I have gained extensive expertise in the analytical understanding and numerical simulation of these systems for both space- and ground-based instruments, achieving unprecedented performance on experimental testbeds and on-sky. Additionally, I am deeply engaged in exosystem science, having analyzed coronagraphic data from various high-contrast instruments. My research complementary integrates instrumental development and observational studies. As coronagraphic instruments grow in complexity, maximizing their potential requires instrumentation expertise, and the insights gained from on-sky observations are essential for shaping future instrumental advancements.

**62 accepted papers in peer-reviewed journals, including:**

- **7 papers as first author + 1 as co-first author,**
- **10 papers with major contributions (2nd or 3rd author),**
- **44 papers with more minor contributions.**

**52 conference proceedings (mainly SPIE), including 11 as first author.**

**All my papers and proceedings are in open access.**

**Google Scholar analytics: 3735 citations – h-index = 34.**

## MAJOR REFEREED PUBLICATIONS

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1. Squicciarini, V. ; **Mazoyer, J.** ; Wilkinson, C. et al. (2025), *GPI+SPHERE detection of a 6.1  $M_{Jup}$  circumbinary planet around HD 143811*, Astronomy and Astrophysics, 702, L10, [DOI link](#), [arXiv link](#), 6 citations
2. Stasevic, S. ; Milli, J. ; **Mazoyer, J.** et al. (2025), *Optimising reference library selection for reference-star differential imaging of discs with SPHERE/IRDIS*, Astronomy and Astrophysics, 701, A93, [DOI link](#), [arXiv link](#)
3. Lajinja, I. ; Baudoz, P. ; **Mazoyer, J.** et al. (2025), *Extended linearity in the high-order wavefront sensor for the Roman Coronagraph*, Astronomy and Astrophysics, 698, A130, [DOI link](#), [arXiv link](#), 4 citations
4. Squicciarini, V. ; **Mazoyer, J.** ; Lagrange, A. -M. et al. (2025), *The COBREX archival survey: Improved constraints on the occurrence rate of wide-orbit substellar companions: I. A uniform re-analysis of 400 stars from the GPIES survey*, Astronomy and Astrophysics, 693, A54, [DOI link](#), [arXiv link](#), 13 citations
5. Gutierrez, Y. ; **Mazoyer, J.** ; Mugnier, L. M. et al. (2024), *Image-based wavefront correction using model-free reinforcement learning*, Optics Express, 32, 31247, [DOI link](#), [arXiv link](#), 2 citations
6. Galicher, R. ; Potier, A. ; **Mazoyer, J.** et al. (2024), *Increasing the raw contrast of VLT/SPHERE with the dark hole technique. III. Broadband reference differential imaging of HR4796 using a four-quadrant phase mask*, Astronomy and Astrophysics, 686, A54, [DOI link](#), [arXiv link](#), 5 citations
7. Galicher, R. & **Mazoyer, J.** (2024), *Imaging exoplanets with coronagraphic instruments*, Comptes Rendus Physique, 24, 133, [DOI link](#), [arXiv link](#), 23 citations
8. Stasevic, S. ; Milli, J. ; **Mazoyer, J.** et al. (2023), *An inner warp discovered in the disk around HD 110058 using VLT/SPHERE and HST/STIS*, Astronomy and Astrophysics, 678, A8, [DOI link](#), [arXiv link](#), 10 citations
9. Potier, A. ; **Mazoyer, J.** ; Wahhaj, Z. et al. (2022), *Increasing the raw contrast of VLT/SPHERE with the dark hole technique. II. On-sky wavefront correction and coherent differential imaging*, Astronomy and Astrophysics, 665, A136, [DOI link](#), [arXiv link](#), 27 citations
10. Chen, C. ; **Mazoyer, J.** ; Poteet, C. A. et al. (2020), *Multiband GPI Imaging of the HR 4796A Debris Disk*, The Astrophysical Journal, 898, 55, [DOI link](#), [arXiv link](#), 41 citations

11. **Mazoyer, J.** ; Pueyo, L. ; N'Diaye, M. et al. (2018), *Active Correction of Aperture Discontinuities-Optimized Stroke Minimization. II. Optimization for Future Missions*, The Astronomical Journal, 155, 8, [DOI link](#), [arXiv link](#), 23 citations
12. **Mazoyer, J.** ; Pueyo, L. ; N'Diaye, M. et al. (2018), *Active Correction of Aperture Discontinuities-Optimized Stroke Minimization. I. A New Adaptive Interaction Matrix Algorithm*, The Astronomical Journal, 155, 7, [DOI link](#), [arXiv link](#), 18 citations
13. Fogarty, K. ; Pueyo, L. ; **Mazoyer, J.** et al. (2017), *Polynomial Apodizers for Centrally Obscured Vortex Coronagraphs*, The Astronomical Journal, 154, 240, [DOI link](#), [arXiv link](#), 10 citations
14. **Mazoyer, J.** ; Pueyo, L. ; Norman, C. et al. (2016), *Active compensation of aperture discontinuities for WFIRST-AFTA: analytical and numerical comparison of propagation methods and preliminary results with a WFIRST-AFTA-like pupil*, Journal of Astronomical Telescopes, Instruments, and Systems, 2, 011008, [DOI link](#), [arXiv link](#), 9 citations
15. **Mazoyer, J.** ; Boccaletti, A. ; Choquet, É. et al. (2016), *A Symmetric Inner Cavity in the HD 141569A Circumstellar Disk*, The Astrophysical Journal, 818, 150, [DOI link](#), [arXiv link](#), 14 citations
16. **Mazoyer, J.** ; Boccaletti, A. ; Augereau, J. -C. et al. (2014), *Is the HD 15115 inner disk really asymmetrical?*, Astronomy and Astrophysics, 569, A29, [DOI link](#), [arXiv link](#), 35 citations
17. **Mazoyer, J.** ; Baudoz, P. ; Galicher, R. et al. (2014), *High-contrast imaging in polychromatic light with the self-coherent camera*, Astronomy and Astrophysics, 564, L1, [DOI link](#), [arXiv link](#), 36 citations
18. **Mazoyer, J.** ; Baudoz, P. ; Galicher, R. et al. (2013), *Estimation and correction of wavefront aberrations using the self-coherent camera: laboratory results*, Astronomy and Astrophysics, 557, A9, [DOI link](#), [arXiv link](#), 41 citations

## OTHER REFEREED PUBLICATIONS

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1. Landman, R. ; Doelman, D. ; Rietjens, J. et al. (2026), *SUPPPRESS: Prototyping and testing liquid-crystal vector vortex coronagraphs with reduced polarization leakage*, accepted in JATIS, [arXiv link](#)
2. Goulas, C. ; Galicher, R. ; Vidal, F. et al. (2026), *Non-common path aberration compensation and a dark hole loop with a pyramid adaptive optics system: Application to SAXO+*, Astronomy and Astrophysics, 708, A50, [DOI link](#), [arXiv link](#)
3. Potier, A. ; Galicher, R. ; Baudoz, P. et al. (2025), *Coherent differential imaging of high-contrast extended sources with VLT/SPHERE*, Astronomy and Astrophysics, 704, A98, [DOI link](#), [arXiv link](#)
4. Hom, J. ; Esposito, T. M. ; Crotts, K. A. et al. (2025), *The Disks In Scorpius-Centaurus Survey (DISCS). I. Four Newly Resolved Debris Disks in Polarized Intensity Light*, The Astronomical Journal, 170, 46, [DOI link](#), [arXiv link](#), 2 citations
5. Lagrange, A. -M. ; Wilkinson, C. ; Mâlin, M. et al. (2025), *Evidence for a sub-Jovian planet in the young TWA 7 disk*, Nature, 642, 905, [DOI link](#), [arXiv link](#), 36 citations
6. Desgrange, C. ; Milli, J. ; Chauvin, G. et al. (2025), *Dust populations from 30 to 1000 au in the debris disk of HD 120326: Panchromatic view with VLT/SPHERE, ALMA, and HST/STIS*, Astronomy and Astrophysics, 698, A183, [DOI link](#), [arXiv link](#), 5 citations
7. Chomez, A. ; Delorme, P. ; Lagrange, A. -M. et al. (2025), *The SPHERE infrared survey for exoplanets (SHINE): IV. Complete observations, data reduction and analysis, detection performances, and final results*, Astronomy and Astrophysics, 697, A99, [DOI link](#), [arXiv link](#), 11 citations
8. Ray, S. ; Sallum, S. ; Hinkley, S. et al. (2025), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. III. Aperture Masking Interferometric Observations of the Star HIP 65426 at 3.8  $\mu\text{m}$* , The Astrophysical Journal, 983, L25, [DOI link](#), [arXiv link](#), 10 citations
9. Lajinja, I. ; Carrión-González, Ó. ; Laugier, R. et al. (2025), *Advancing European high-contrast imaging R&D towards the Habitable Worlds Observatory*, Astrophysics and Space Science, 370, 29, [DOI link](#), [arXiv link](#), 1 citation

10. Wilkinson, C. ; Charnay, B. ; Mazevet, S. et al. (2024), *Breaking degeneracies in exoplanetary parameters through self-consistent atmosphere-interior modelling*, *Astronomy and Astrophysics*, 692, A113, [DOI link](#), [arXiv link](#), 12 citations
11. Lewis, B. L. ; Fitzgerald, M. P. ; Esposito, T. M. et al. (2024), *Gemini Planet Imager Observations of a Resolved Low-inclination Debris Disk around HD 156623*, *The Astronomical Journal*, 168, 142, [DOI link](#), [arXiv link](#), 2 citations
12. Goulas, C. ; Galicher, R. ; Vidal, F. et al. (2024), *Numerical simulations for the SAXO+ upgrade: Performance analysis of the adaptive optics system*, *Astronomy and Astrophysics*, 689, A199, [DOI link](#), [arXiv link](#), 5 citations
13. Petrus, S. ; Whiteford, N. ; Patapis, P. et al. (2024), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. V. Do Self-consistent Atmospheric Models Represent JWST Spectra? A Showcase with VHS 1256-1257 b*, *The Astrophysical Journal*, 966, L11, [DOI link](#), [arXiv link](#), 42 citations
14. Hom, J. ; Patience, J. ; Chen, C. H. et al. (2024), *A uniform analysis of debris discs with the Gemini Planet Imager II: constraints on dust density distribution using empirically informed scattering phase functions*, *Monthly Notices of the Royal Astronomical Society*, 528, 6959, [DOI link](#), [arXiv link](#), 11 citations
15. Sallum, S. ; Ray, S. ; Kammerer, J. et al. (2024), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. IV. NIRISS Aperture Masking Interferometry Performance and Lessons Learned*, *The Astrophysical Journal*, 963, L2, [DOI link](#), [arXiv link](#), 13 citations
16. Worthen, K. ; Chen, C. H. ; Brittain, S. D. et al. (2024), *Vertical Structure of Gas and Dust in Four Debris Disks*, *The Astrophysical Journal*, 962, 166, [DOI link](#), [arXiv link](#), 6 citations
17. Crotts, K. A. ; Matthews, B. C. ; Duchêne, G. et al. (2024), *A Uniform Analysis of Debris Disks with the Gemini Planet Imager. I. An Empirical Search for Perturbations from Planetary Companions in Polarized Light Images*, *The Astrophysical Journal*, 961, 245, [DOI link](#), [arXiv link](#), 19 citations
18. Vaughan, S. R. ; Gebhard, T. D. ; Bott, K. et al. (2023), *Chasing rainbows and ocean glints: Inner working angle constraints for the Habitable Worlds Observatory*, *Monthly Notices of the Royal Astronomical Society*, 524, 5477, [DOI link](#), [arXiv link](#), 48 citations
19. Carter, A. L. ; Hinkley, S. ; Kammerer, J. et al. (2023), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems I: High-contrast Imaging of the Exoplanet HIP 65426 b from 2 to 16  $\mu\text{m}$* , *The Astrophysical Journal*, 951, L20, [DOI link](#), [arXiv link](#), 113 citations
20. Miles, B. E. ; Biller, B. A. ; Patapis, P. et al. (2023), *The JWST Early-release Science Program for Direct Observations of Exoplanetary Systems II: A 1 to 20  $\mu\text{m}$  Spectrum of the Planetary-mass Companion VHS 1256-1257 b*, *The Astrophysical Journal*, 946, L6, [DOI link](#), [arXiv link](#), 178 citations
21. Hinkley, S. ; Carter, A. L. ; Ray, S. et al. (2022), *The JWST Early Release Science Program for the Direct Imaging and Spectroscopy of Exoplanetary Systems*, *Publications of the Astronomical Society of the Pacific*, 134, 095003, [DOI link](#), [arXiv link](#), 57 citations
22. Crotts, K. A. ; Draper, Z. H. ; Matthews, B. C. et al. (2022), *A Multiwavelength Study of the Highly Asymmetrical Debris Disk around HD 111520*, *The Astrophysical Journal*, 932, 23, [DOI link](#), [arXiv link](#), 8 citations
23. Betti, S. K. ; Follette, K. ; Jorquera, S. et al. (2022), *Detection of Near-infrared Water Ice at the Surface of the (Pre)Transitional Disk of AB Aur: Informing Icy Grain Abundance, Composition, and Size*, *The Astronomical Journal*, 163, 145, [DOI link](#), [arXiv link](#), 20 citations
24. Singh, G. ; Bhowmik, T. ; Boccaletti, A. et al. (2021), *Revealing asymmetrical dust distribution in the inner regions of HD 141569*, *Astronomy and Astrophysics*, 653, A79, [DOI link](#), [arXiv link](#), 15 citations
25. Crotts, K. A. ; Matthews, B. C. ; Esposito, T. M. et al. (2021), *A Deep Polarimetric Study of the Asymmetrical Debris Disk HD 106906*, *The Astrophysical Journal*, 915, 58, [DOI link](#), [arXiv link](#), 16 citations
26. Arriaga, P. ; Fitzgerald, M. P. ; Duchêne, G. et al. (2020), *Multiband Polarimetric Imaging of HR 4796A with the Gemini Planet Imager*, *The Astronomical Journal*, 160, 79, [DOI link](#), [arXiv link](#), 35 citations

27. Esposito, T. M. ; Kalas, P. ; Fitzgerald, M. P. et al. (2020), *Debris Disk Results from the Gemini Planet Imager Exoplanet Survey's Polarimetric Imaging Campaign*, The Astronomical Journal, 160, 24, [DOI link](#), [arXiv link](#), 110 citations
28. Duchêne, G. ; Rice, M. ; Hom, J. et al. (2020), *The Gemini Planet Imager View of the HD 32297 Debris Disk*, The Astronomical Journal, 159, 251, [DOI link](#), [arXiv link](#), 27 citations
29. Ertel, S. ; Defrère, D. ; Hinz, P. et al. (2020), *The HOSTS Survey for Exozodiacal Dust: Observational Results from the Complete Survey*, The Astronomical Journal, 159, 177, [DOI link](#), [arXiv link](#), 133 citations
30. Bruzzone, J. S. ; Metchev, S. ; Duchêne, G. et al. (2020), *Imaging the 44 au Kuiper Belt Analog Debris Ring around HD 141569A with GPI Polarimetry*, The Astronomical Journal, 159, 53, [DOI link](#), [arXiv link](#), 12 citations
31. Hom, J. ; Patience, J. ; Esposito, T. M. et al. (2020), *First Resolved Scattered-light Images of Four Debris Disks in Scorpius-Centaurus with the Gemini Planet Imager*, The Astronomical Journal, 159, 31, [DOI link](#), [arXiv link](#), 16 citations
32. Bhowmik, T. ; Boccaletti, A. ; Thébault, P. et al. (2019), *Spatially resolved spectroscopy of the debris disk HD 32297. Further evidence of small dust grains*, Astronomy and Astrophysics, 630, A85, [DOI link](#), [arXiv link](#), 38 citations
33. Ren, B. ; Choquet, É. ; Perrin, M. D. et al. (2019), *An Exo-Kuiper Belt with an Extended Halo around HD 191089 in Scattered Light*, The Astrophysical Journal, 882, 64, [DOI link](#), [arXiv link](#), 43 citations
34. Stark, C. C. ; Belikov, R. ; Bolcar, M. R. et al. (2019), *ExoEarth yield landscape for future direct imaging space telescopes*, Journal of Astronomical Telescopes, Instruments, and Systems, 5, 024009, [DOI link](#), [arXiv link](#), 83 citations
35. Engler, N. ; Boccaletti, A. ; Schmid, H. M. et al. (2019), *Investigating the presence of two belts in the HD 15115 system*, Astronomy and Astrophysics, 622, A192, [DOI link](#), [arXiv link](#), 35 citations
36. Esposito, T. M. ; Duchêne, G. ; Kalas, P. et al. (2018), *Direct Imaging of the HD 35841 Debris Disk: A Polarized Dust Ring from Gemini Planet Imager and an Outer Halo from HST/STIS*, The Astronomical Journal, 156, 47, [DOI link](#), [arXiv link](#), 35 citations
37. Leboulleux, L. ; Sauvage, J. -F. ; Pueyo, L. A. et al. (2018), *Pair-based Analytical model for Segmented Telescopes Imaging from Space for sensitivity analysis*, Journal of Astronomical Telescopes, Instruments, and Systems, 4, 035002, [DOI link](#), [arXiv link](#), 21 citations
38. Poteet, C. A. ; Chen, C. H. ; Hines, D. C. et al. (2018), *Space-based Coronagraphic Imaging Polarimetry of the TW Hydrae Disk: Shedding New Light on Self-shadowing Effects*, The Astrophysical Journal, 860, 115, [DOI link](#), [arXiv link](#), 13 citations
39. Jensen-Clem, R. ; Mawet, D. ; Gomez Gonzalez, C. A. et al. (2018), *A New Standard for Assessing the Performance of High Contrast Imaging Systems*, The Astronomical Journal, 155, 19, [DOI link](#), [arXiv link](#), 34 citations
40. Perrot, C. ; Boccaletti, A. ; Pantin, E. et al. (2016), *Discovery of concentric broken rings at sub-arcsec separations in the HD 141569A gas-rich, debris disk with VLT/SPHERE*, Astronomy and Astrophysics, 590, L7, [DOI link](#), [arXiv link](#), 47 citations
41. Delorme, J. R. ; Galicher, R. ; Baudoz, P. et al. (2016), *Focal plane wavefront sensor achromatization: The multireference self-coherent camera*, Astronomy and Astrophysics, 588, A136, [DOI link](#), [arXiv link](#), 19 citations
42. Choquet, É. ; Perrin, M. D. ; Chen, C. H. et al. (2016), *First Images of Debris Disks around TWA 7, TWA 25, HD 35650, and HD 377*, The Astrophysical Journal, 817, L2, [DOI link](#), [arXiv link](#), 87 citations
43. Debes, J. H. ; Ygouf, M. ; Choquet, E. et al. (2016), *Wide-Field Infrared Survey Telescope-Astrophysics Focused Telescope Assets coronagraphic operations: lessons learned from the Hubble Space Telescope and the James Webb Space Telescope*, Journal of Astronomical Telescopes, Instruments, and Systems, 2, 011010, [DOI link](#), [arXiv link](#), 12 citations

44. Wiens, R. C. ; Maurice, S. ; Lasue, J. et al. (2013), *Pre-flight calibration and initial data processing for the ChemCam laser-induced breakdown spectroscopy instrument on the Mars Science Laboratory rover*, Spectrochimica Acta - Part B: Atomic Spectroscopy, 82, 1, [DOI link](#), 180 citations
45. Cousin, A. ; Forni, O. ; Maurice, S. et al. (2011), *Laser induced breakdown spectroscopy library for the Martian environment*, Spectrochimica Acta - Part B: Atomic Spectroscopy, 66, 805, [DOI link](#), 63 citations

## MAJOR CONFERENCE PROCEEDINGS

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1. Mazoyer, J. ; Mellado-Tenorio, M. J. ; Potier, A. et al. (2026), *Coherence differential imaging using gradient-boosted decision trees for the direct detection of exoplanets*, Space Telescopes and Instrumentation 2026: Optical, Infrared, and Millimeter Wave, [arXiv link](#)
2. Mazoyer, J. ; Goulas, C. ; Galicher, R. et al. (2026), *SAXO+, the second-stage adaptive optics for SPHERE: NCPA compensation and dark-hole loop with a pyramid wavefront sensor*, Adaptive Optics Systems X, [DOI link](#), [arXiv link](#)
3. Gutierrez, Y. ; Mazoyer, J. ; Herscovici-Schiller, O. et al. (2024), *A deep reinforcement learning approach to wavefront control for exoplanet imaging*, Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave, 13092, 130926H, [DOI link](#), [arXiv link](#), 2 citations
4. Mazoyer, J. ; Goulas, C. ; Vidal, F. et al. (2024), *Upgrading SPHERE with the second stage AO system SAXO+: non-common path aberrations estimation and correction*, Ground-based and Airborne Instrumentation for Astronomy X, 13096, 130969D, [DOI link](#)
5. Fogarty, K. ; Mawet, D. ; Mazoyer, J. et al. (2020), *Towards high throughput and low-order aberration robustness for vortex coronagraphs with central obstructions*, Space Telescopes and Instrumentation 2020: Optical, Infrared, and Millimeter Wave, 11443, 114433Y, [DOI link](#), 1 citation
6. Mazoyer, J. ; Arriaga, P. ; Hom, J. et al. (2020), *DiskFM: A forward modeling tool for disk analysis with coronagraphic instruments*, Ground-based and Airborne Instrumentation for Astronomy VIII, 11447, 1144759, [DOI link](#), [arXiv link](#), 11 citations
7. Fogarty, K. ; Mazoyer, J. ; St. Laurent, K. et al. (2018), *Optimal deformable mirror and pupil apodization combinations for apodized pupil Lyot coronagraphs with obstructed pupils*, Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave, 10698, 106981J, [DOI link](#), 2 citations
8. Ruane, G. ; Riggs, A. ; Mazoyer, J. et al. (2018), *Review of high-contrast imaging systems for current and future ground- and space-based telescopes I: coronagraph design methods and optical performance metrics*, Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave, 10698, 106982S, [DOI link](#), [arXiv link](#), 17 citations
9. Mazoyer, J. ; Pueyo, L. ; N'Diaye, M. et al. (2017), *Capabilities of ACAD-OSM, an active method for the correction of aperture discontinuities*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 10400, 104000G, [DOI link](#), [arXiv link](#), 2 citations
10. Mazoyer, J. & Pueyo, L. (2017), *Fundamental limits to high-contrast wavefront control*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 10400, 1040014, [DOI link](#), [arXiv link](#), 4 citations
11. Leboulleux, L. ; N'Diaye, M. ; Mazoyer, J. et al. (2017), *Comparison of wavefront control algorithms and first results on the high-contrast imager for complex aperture telescopes (hicat) testbed*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 10562, 105622Z, [DOI link](#)
12. Fogarty, K. ; Pueyo, L. ; Mazoyer, J. et al. (2017), *Tip/tilt optimizations for polynomial apodized vortex coronagraphs on obscured telescope pupils*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 10400, 104000T, [DOI link](#), 2 citations
13. Mazoyer, J. ; Pueyo, L. ; N'Diaye, M. et al. (2016), *Correcting for the effects of pupil discontinuities with the ACAD method*, Space Telescopes and Instrumentation 2016: Optical, Infrared, and Millimeter Wave, 9904, 99044T, [DOI link](#), [arXiv link](#), 2 citations

14. **Mazoyer, J.** ; Pueyo, L. ; Norman, C. et al. (2015), *Active compensation of aperture discontinuities for WFIRST- AFTA: analytical and numerical comparison of propagation methods and preliminary results with an AFTA like pupil*, Nancy Grace Roman Space Telescope Technical Report WFIRST-STScI-TR1505, WFIRST-STScI-TR1505
15. **Mazoyer, J.** ; Pueyo, L. ; Norman, C. et al. (2015), *Active correction of aperture discontinuities (ACAD) for space telescope pupils: a parametric analysis*, Techniques and Instrumentation for Detection of Exoplanets VII, 9605, 96050M, [DOI link](#), [arXiv link](#), 10 citations
16. N'Diaye, M. ; **Mazoyer, J.** ; Choquet, É. et al. (2015), *High-contrast imager for complex aperture telescopes (HiCAT): 3. first lab results with wavefront control*, Techniques and Instrumentation for Detection of Exoplanets VII, 9605, 96050I, [DOI link](#), [arXiv link](#), 23 citations
17. **Mazoyer, J.** ; Galicher, R. ; Baudoz, P. et al. (2014), *Deformable mirror interferometric analysis for the direct imagery of exoplanets*, Adaptive Optics Systems IV, 9148, 914846, [DOI link](#), [arXiv link](#), 3 citations
18. **Mazoyer, J.** ; Baudoz, P. ; Galicher, R. et al. (2013), *Direct detection of exoplanets in polychromatic light with a Self-coherent camera*, Proceedings of the Third AO4ELT Conference, 97, [DOI link](#)
19. Baudoz, P. ; **Mazoyer, J.** ; Galicher, R. (2013), *Laboratory tests of planet signal extraction in high contrast images*, Proceedings of the Third AO4ELT Conference, 109, [DOI link](#), 2 citations
20. **Mazoyer, J.** ; Galicher, R. ; Baudoz, P. et al. (2013), *Speckle correction in polychromatic light with the self-coherent camera for the direct detection of exoplanets*, Techniques and Instrumentation for Detection of Exoplanets VI, 8864, 88640N, [DOI link](#), [arXiv link](#), 1 citation
21. Galicher, R. ; **Mazoyer, J.** ; Baudoz, P. et al. (2013), *High-contrast imaging with a self-coherent camera*, Techniques and Instrumentation for Detection of Exoplanets VI, 8864, 88640M, [DOI link](#)
22. Baudoz, P. ; **Mazoyer, J.** ; Mas, M. et al. (2012), *Dark hole and planet detection: laboratory results using the self-coherent camera*, Ground-based and Airborne Instrumentation for Astronomy IV, 8446, 84468C, [DOI link](#), 11 citations
23. Mas, M. ; Baudoz, P. ; **Mazoyer, J.** et al. (2012), *Experimental results on wavefront correction using the self-coherent camera*, Ground-based and Airborne Instrumentation for Astronomy IV, 8446, 844689, [DOI link](#), 4 citations
24. **Mazoyer, J.** ; Baudoz, P. ; Mas, M. et al. (2012), *Experimental parametric study of the self-coherent camera*, Space Telescopes and Instrumentation 2012: Optical, Infrared, and Millimeter Wave, 8442, 844250, [DOI link](#), [arXiv link](#), 2 citations
25. Gasnault, O. ; **Mazoyer, J.** ; Cousin, A. et al. (2012), *Deciphering Sample and Atmospheric Oxygen Contents with ChemCam on Mars*, 43rd Annual Lunar and Planetary Science Conference, 2888, 1 citation

## OTHER CONFERENCE PROCEEDINGS

---

1. Stadler, E. ; Schreiber, L. ; Cortecchia, F. et al. (2024), *Upgrading SPHERE with the second-stage adaptive optics system SAXO+: conceptual design of the opto-mechanical module*, Adaptive Optics Systems IX, 13097, 130976S, [DOI link](#), 1 citation
2. Goulas, C. ; Galicher, R. ; Vidal, F. et al. (2024), *Upgrading SPHERE with the second stage AO system SAXO+: exploration of the parameter space with end-to-end numerical simulations*, Adaptive Optics Systems IX, 13097, 1309769, [DOI link](#)
3. Cantalloube, F. ; Christiaens, V. ; Cantero, C. et al. (2024), *Exoplanet imaging data challenge, phase II: comparison of algorithms in terms of characterization capabilities*, Adaptive Optics Systems IX, 13097, 1309713, [DOI link](#), [arXiv link](#)
4. Potier, A. ; Riggs, A. J. E. ; Ruane, G. et al. (2024), *Revisiting the Borde-Traub focal plane wavefront estimation technique for exoplanet direct imaging*, Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave, 13092, 130926E, [DOI link](#)

5. Savransky, D. ; Bailey, V. P. ; Wolff, S. G. et al. (2024), *The Nancy Grace Roman Space Telescope coronagraph community participation program*, Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave, 13092, 130921I, [DOI link](#), 2 citations
6. Millar-Blanchaer, M. A. ; Wang, J. ; Bogat, E. et al. (2024), *The Roman coronagraph community participation program: data reduction and simulations*, Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave, 13092, 1309256, [DOI link](#), 1 citation
7. Betti, S. ; Chen, C. ; Beck, T. et al. (2024), *Coronagraphic JWST/NIRCam Images of the 49 Ceti Debris Disk*, AASTCS10, Extreme Solar Systems V, 56, 623.04
8. Hom, J. ; Douglas, E. ; Anche, R. et al. (2024), *Searching for Signs of Earth-Like Planets: Demonstrating the Potential of the Roman-Coronagraph in Resolving Exozodiacal Dust and Debris Disk Substructures*, AASTCS10, Extreme Solar Systems V, 56, 623.02
9. Fowler, J. ; Haffert, S. Y. ; van Kooten, M. A. M. et al. (2023), *Visible extreme adaptive optics on extremely large telescopes: towards detecting oxygen in Proxima Centauri b and analogs*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 12680, 126801U, [DOI link](#), [arXiv link](#), 3 citations
10. Desai, N. ; König, L. ; Por, E. et al. (2023), *Integrated photonic-based coronagraphic systems for future space telescopes*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 12680, 126801S, [DOI link](#), [arXiv link](#), 1 citation
11. Béchet, C. ; Tallon, M. ; Thiébaud, E. et al. (2023), *Inverse problem approach for SPHERE+ adaptive optics control*, Adaptive Optics for Extremely Large Telescopes (AO4ELT7), 47, [DOI link](#)
12. Stadler, E. ; Diolaiti, E. ; Schreiber, L. et al. (2023), *Status report of the SAXO+ opto-mechanical design concept*, Adaptive Optics for Extremely Large Telescopes (AO4ELT7), 127, [DOI link](#)
13. Goulas, C. ; Vidal, F. ; Galicher, R. et al. (2023), *SAXO+ upgrade : second stage AO system end-to-end numerical simulations*, Adaptive Optics for Extremely Large Telescopes (AO4ELT7), 32, [DOI link](#)
14. Cantalloube, F. ; Christiaens, V. ; Cantero, C. et al. (2022), *Exoplanet imaging data challenge, phase II: characterization of exoplanet signals in high-contrast images*, Adaptive Optics Systems VIII, 12185, 1218505, [DOI link](#), [arXiv link](#), 2 citations
15. Boccaletti, A. ; Chauvin, G. ; Wildi, F. et al. (2022), *Upgrading the high contrast imaging facility SPHERE: science drivers and instrument choices*, Ground-based and Airborne Instrumentation for Astronomy IX, 12184, 121841S, [DOI link](#), [arXiv link](#), 5 citations
16. Potier, A. ; Wahhaj, Z. ; Galicher, R. et al. (2022), *Improving VLT/SPHERE without additional hardware: comparing quasi-static correction strategies*, Adaptive Optics Systems VIII, 12185, 1218568, [DOI link](#), [arXiv link](#)
17. N'Diaye, M. ; Fogarty, K. ; Soummer, R. et al. (2018), *Apodized Pupil Lyot coronagraphs with arbitrary aperture telescopes: novel designs using hybrid focal plane masks*, Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave, 10698, 106986A, [DOI link](#), [arXiv link](#), 3 citations
18. Soummer, R. ; Brady, G. R. ; Brooks, K. et al. (2018), *High-contrast imager for complex aperture telescopes (HiCAT): 5. first results with segmented-aperture coronagraph and wavefront control*, Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave, 10698, 106981O, [DOI link](#), [arXiv link](#), 10 citations
19. Snik, F. ; Absil, O. ; Baudoz, P. et al. (2018), *Review of high-contrast imaging systems for current and future ground-based and space-based telescopes III: technology opportunities and pathways*, Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation III, 10706, 107062L, [DOI link](#), [arXiv link](#), 4 citations
20. St. Laurent, K. ; Fogarty, K. ; Zimmerman, N. T. et al. (2018), *Apodized pupil Lyot coronagraphs designs for future segmented space telescopes*, Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave, 10698, 106982W, [DOI link](#), [arXiv link](#), 4 citations
21. Jovanovic, N. ; Absil, O. ; Baudoz, P. et al. (2018), *Review of high-contrast imaging systems for current and future ground-based and space-based telescopes: Part II. Common path wavefront sensing/control*

- and coherent differential imaging*, Adaptive Optics Systems VI, 10703, 107031U, [DOI link](#), [arXiv link](#), 18 citations
22. Leboulleux, L. ; Pueyo, L. ; Sauvage, J. -F. et al. (2018), *Sensitivity analysis for high-contrast imaging with segmented space telescopes*, Space Telescopes and Instrumentation 2018: Optical, Infrared, and Millimeter Wave, 10698, 106986H, [DOI link](#), 3 citations
  23. Egrou, S. ; Soummer, R. ; Lajoie, C. -P. et al. (2017), *James Webb Space Telescope optical simulation testbed IV: linear control alignment of the primary segmented mirror*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 10398, 1039811, [DOI link](#)
  24. Pueyo, L. ; Zimmerman, N. ; Bolcar, M. et al. (2017), *The LUVOIR architecture "A" coronagraph instrument*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, 10398, 103980F, [DOI link](#), 11 citations
  25. Leboulleux, L. ; N'Diaye, M. ; Riggs, A. J. E. et al. (2016), *High-contrast imager for Complex Aperture Telescopes (HiCAT). 4. Status and wavefront control development*, Space Telescopes and Instrumentation 2016: Optical, Infrared, and Millimeter Wave, 9904, 99043C, [DOI link](#), 10 citations
  26. Debes, J. H. ; Ygouf, M. ; Choquet, E. et al. (2015), *WFIRST-AFTA Coronagraphic Operations: Lessons Learned from the Hubble Space Telescope and the James Webb Space Telescope*, Nancy Grace Roman Space Telescope Technical Report WFIRST-STScI-TR1504, WFIRST-STScI-TR1504
  27. Galicher, R. ; Baudoz, P. ; Delorme, J. R. et al. (2014), *High contrast imaging on the THD bench: progress and upgrades*, Space Telescopes and Instrumentation 2014: Optical, Infrared, and Millimeter Wave, 9143, 91435A, [DOI link](#), 3 citations
  28. Delorme, J. R. ; Galicher, R. ; Baudoz, P. et al. (2014), *High-contrast imaging in wide spectral band with a self-coherent camera and achromatic coronagraphs*, Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation, 9151, 91515Q, [DOI link](#), 1 citation
  29. Galicher, R. ; Delorme, J. R. ; Baudoz, P. et al. (2013), *Focal Plane Wavefront Sensing with a self-coherent camera*, Proceedings of the Third AO4ELT Conference, 123, [DOI link](#)

## WHITE PAPERS (SELECTED)

---

- Boccaletti, A. et al. (2020), *SPHERE+: Imaging young Jupiters down to the snowline*, arXiv e-prints, [arXiv:2003.05714](#)
- Gaudi, B. S. et al. (2020), *The Habitable Exoplanet Observatory (HabEx) Mission Concept Study Final Report*, arXiv e-prints, [arXiv:2001.06683](#)
- The LUVOIR Team (2019), *The LUVOIR Mission Concept Study Final Report*, arXiv e-prints, [arXiv:1912.06219](#)
- Mazoyer, J. et al. (2019), *High-Contrast Testbeds for Future Space-Based Direct Imaging Exoplanet Missions*, Bulletin of the American Astronomical Society, 51, 101, [arXiv:1907.09508](#)

## THESIS

---

- Mazoyer, J. (2024) Optique active pour l'imagerie d'exoplanètes et de disques de débris. Thèse d'habilitation. Observatoire de Paris - PSL, [HAL link](#) - [Defense Youtube link](#)
- Mazoyer, J. (2014) Haut contraste pour l'imagerie directe d'exoplanètes et de disques : de la self-coherent camera à l'analyse de données NICI. Thèse de doctorat. Université Paris Diderot - Paris 7, [DOI link](#)